

A Statistical Mechanical Analysis of the Ising Model for Beginners

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January 4, 2026

1 Introduction

This document contains my personal refraction of paramagnetic and 1D Ising model thermodynamics, completed as part of a statistical mechanics course. Although I worked through the mathematics independently, the problems, methods, and results are part of the standard physics curriculum dating back to Ising's original 1925 work.

I started with the simple case: non-interacting spins in a magnetic field (paramagnetism). Then I added spin-spin interactions (ferromagnetism but through a simpler mean-field effect), working through the 1D Ising model step by step.

All derivations are shown explicitly. No fancy jargon, no long-winded explanations—just the math as I worked it out. I am sharing this online in case someone else is trying to understand the basics.

The calculations are mine, done for a group project in Statistical Mechanics. Although I cannot guaranty the credibility of the work to 100% as I am a student myself, my professor found no conceptual problems with my approach, though we discussed that the ferromagnetic treatment uses an independent-bond approximation.

Note on the ferromagnetic treatment: While the paramagnetic analysis is exact, my treatment of ferromagnetism uses a simplified factorization that assumes independent nearest-neighbor bonds. This corresponds to a mean-field-like approximation where each spin-spin interaction is treated as statistically independent. While this captures qualitative features (like the absence of finite-temperature phase transitions in 1D), it differs from the exact 1D Ising model solution obtained via the transfer matrix method. I've chosen this approach for its pedagogical clarity and algebraic simplicity.

Physically, this independent-bond approximation treats each neighbor pair as if it were isolated from the rest of the chain, neglecting correlations that propagate along the chain in the exact Ising model.

This simplification was also necessary to describe the system I simulated with my code. I believe my professor mentioned that the involvement of matrices would make the algebra harder. Further, there are several resources already available for tackling the math of the exact Ising Model. At an undergraduate level, considering simplifications that are necessary, this is what I ended up with.

2 Mathematical Analysis (Paramagnetism)

To model the system mathematically, we initialize the spins of the individual particles to be oriented randomly "up" or "down." Conventionally, up is positive and down is negative. So, if we subject the paramagnetic or ferromagnetic material to a magnetic field, $\mathbf{B}$, we expect spins to align with field (flip to +1 if they were initially -1) depending on the strength of the field compared to the temperatyre of the system.

2.1 Partition Function, Z

We start with the partition function,

$$Z = e^{-\beta E_1} + e^{\beta E_2}$$

The energy of our states is given by the definition of the potential energy of a magnetic dipole:

$$E = -\mu \cdot \mathbf{B}$$

Plugging this into our partition function, we get,

$$Z = e^{-\beta\mu B} + e^{-\beta\mu B}$$

A trigonometric identity helps us transform this into a hyperbolic cos (cosh) function for easier handling:

$$Z = 2 \cosh(\beta\mu B)$$

For N independent spins, this is,

$$Z_N = (2 \cosh(\beta\mu B))^N$$

2.2 Magnetization

The magnetization of a single spin is given by its magnetic moment:

$$m = \mu\sigma = \begin{cases} +\mu & \text{for spin up } (\sigma = +1) \\ -\mu & \text{for spin down } (\sigma = -1) \end{cases}$$

The average magnetization is the statistical expectation value:

$$\langle m \rangle = \sum_{\sigma=\pm 1} (\mu\sigma) \cdot P(\sigma)$$

where the probability of each state is given by the Boltzmann distribution:

$$P(\sigma) = \frac{e^{-\beta E_\sigma}}{Z}$$

Substituting the energies $E_+ = -\mu B$ (spin up) and $E_- = +\mu B$ (spin down):

$$\langle m \rangle = \frac{(+\mu)e^{-\beta(-\mu B)} + (-\mu)e^{-\beta(+\mu B)}}{Z}$$

$$\langle m \rangle = \mu \cdot \frac{e^{\beta\mu B} - e^{-\beta\mu B}}{2 \cosh(\beta\mu B)}$$

Similar to what we did for the partition function, we reduce the numerator to a hyperbolic *sin* this time with the same factor of 2. The 2s are common factors and disappear, we get the hyperbolic tangent:

$$\langle m \rangle = \mu \cdot \tanh(\beta\mu B)$$

For N independent spins, the total magnetization is:

$$M = N \langle m \rangle = N\mu \cdot \tanh(\beta\mu B)$$

2.3 Fluctuations

The variance of the magnetization provides insight into thermal fluctuations:

$$\text{Var}(M) = \langle M^2 \rangle - \langle M \rangle^2$$

For a single spin, the second moment is:

$$\langle m^2 \rangle = \frac{(+\mu)^2 e^{\beta\mu B} + (-\mu)^2 e^{-\beta\mu B}}{e^{\beta\mu B} + e^{-\beta\mu B}}$$

$$\langle m^2 \rangle = \mu^2 \cdot \frac{e^{\beta\mu B} + e^{-\beta\mu B}}{e^{\beta\mu B} + e^{-\beta\mu B}} = \mu^2$$

Thus for one spin:

$$\begin{aligned}\langle m^2 \rangle - \langle m \rangle^2 &= \mu^2 - \mu^2 \tanh^2(\beta\mu B) \\ &= \mu^2 [1 - \tanh^2(\beta\mu B)] = \mu^2 \operatorname{sech}^2(\beta\mu B)\end{aligned}$$

For N independent spins, variances add:

$$\langle M^2 \rangle - \langle M \rangle^2 = N\mu^2 \operatorname{sech}^2(\beta\mu B)$$

Or equivalently:

$$\langle M^2 \rangle - \langle M \rangle^2 = N\mu^2 \operatorname{sech}^2\left(\frac{\mu B}{k_B T}\right)$$

2.4 Free Energy

The Helmholtz free energy is obtained from the partition function:

$$F = -k_B T \ln Z_N$$

Substituting $Z_N = [2 \cosh(\beta\mu B)]^N$:

$$F = -k_B T \ln ([2 \cosh(\beta\mu B)]^N)$$

$$F = -Nk_B T \ln [2 \cosh(\beta\mu B)]$$

Or explicitly:

$$F = -Nk_B T \ln \left[2 \cosh \left(\frac{\mu B}{k_B T} \right) \right]$$

2.5 Energy

The average energy can be obtained from:

$$E = -\frac{\partial \ln Z_N}{\partial \beta}$$

First compute:

$$\ln Z_N = N \ln [2 \cosh(\beta\mu B)]$$

Taking the derivative:

$$\frac{\partial \ln Z_N}{\partial \beta} = N \cdot \frac{\mu B \sinh(\beta\mu B)}{\cosh(\beta\mu B)} = N\mu B \tanh(\beta\mu B)$$

Thus:

$$E = -N\mu B \tanh(\beta\mu B)$$

Or:

$$E = -N\mu B \tanh\left(\frac{\mu B}{k_B T}\right)$$

Note the negative sign: energy decreases as spins align with the field.

2.6 Entropy

The entropy follows from the thermodynamic relation:

$$S = k_B (\ln Z_N + \beta E)$$

Substituting our expressions:

$$S = k_B [N \ln [2 \cosh(\beta\mu B)] + \beta (-N\mu B \tanh(\beta\mu B))]$$

$$S = Nk_B [\ln [2 \cosh(\beta\mu B)] - \beta\mu B \tanh(\beta\mu B)]$$

Using the identity $\ln[2 \cosh(x)] = x \tanh(x) + \ln[1 + e^{-2x}]$, we can also write:

$$S = Nk_B \left[\ln(1 + e^{-2\beta\mu B}) - \frac{2\beta\mu B}{1 + e^{2\beta\mu B}} \right]$$

Or explicitly in terms of temperature:

$$S = Nk_B \left[\ln \left[2 \cosh \left(\frac{\mu B}{k_B T} \right) \right] - \frac{\mu B}{k_B T} \tanh \left(\frac{\mu B}{k_B T} \right) \right]$$

3 Mathematical Analysis (Ferromagnetism)

3.1 Defining the Hamiltonian

For a general ferromagnetic system with spin-spin interactions, we start with the Heisenberg model Hamiltonian:

$$H = -\mu \sum_{i=1}^N \mathbf{B} \cdot \mathbf{S}_i - J \sum_{i=1}^N (\mathbf{S}_i \cdot \mathbf{S}_{i-1} + \mathbf{S}_i \cdot \mathbf{S}_{i+1})$$

where:

- μ is the magnetic moment
- $\mathbf{B}$ is the external magnetic field
- $\mathbf{S}_i$ are quantum spin operators (vectors)
- $J > 0$ is the ferromagnetic coupling constant (favoring parallel alignment)
- The second sum represents nearest-neighbor interactions

The **Ising model** simplifies this drastically by:

1. Restricting spins to one dimension: $\mathbf{S}_i \rightarrow S_i$ (scalar)
2. Allowing only two orientations: $S_i = \pm 1$ (up or down)
3. Considering only nearest-neighbor interactions along a chain

Thus, the 1D Ising model Hamiltonian becomes:

$$H = -\mu B \sum_{i=1}^N S_i - J \sum_{i=1}^{N-1} S_i S_{i+1}$$

where $S_i = \pm 1$, and we've assumed periodic boundary conditions are *not* applied (open chain).

3.2 Partition Function

The canonical partition function is:

$$Z = \sum_{\{S_i\}} e^{-\beta H}$$

where the sum is over all 2^N possible spin configurations: $\{S_i\} = (S_1, S_2, \dots, S_N)$ with each $S_i = \pm 1$.

Substituting the Ising Hamiltonian:

$$Z = \sum_{\{S_i\}} \exp \left[\beta \mu B \sum_{i=1}^N S_i + \beta J \sum_{i=1}^{N-1} S_i S_{i+1} \right]$$

This factorizes into two products:

$$Z = \sum_{\{S_i\}} \left(\prod_{i=1}^N e^{\beta \mu B S_i} \right) \left(\prod_{i=1}^{N-1} e^{\beta J S_i S_{i+1}} \right)$$

We must first take care of some details in this equation before we simplify it. The Heisenberg model handles the coupling constant, J , better than the Ising model in this version.

Considering the Hamiltonian of the Heisenberg model first,

$$Z = \sum_{\{S_i\}} \left(\prod_{i=1}^N e^{\beta \mu B S_i} \right) \left(\prod_{i=1}^{N-1} e^{\beta J (S_i S_{i+1} + S_i S_{i-1})} \right)$$

when we actually perform the summation for a particle i , if the particles beside it have different spins, i.e., when

$$S_{i+1} = S_{i-1}$$

we get a value of

$$-2J \sum_{i=1} S_i S_{i+1}$$

for the second term. And when

$$S_{i+1} \neq S_{i-1}$$

The term becomes 0.

Because the 1D Ising model does not consider the spin of the $i-1$ particle (because of my mean-field-like approximation), we risk not considering the weight of the coupling constant correctly.

To avoid this faulty weight-age of how the coupling constant affects the spin of the particle in question, we define a new term:

$$z = S_i S_{i+1}$$

In this particular version of math, due to the independent-bond assumption, we can stick to the version where

$$S_{i+1} = S_{i-1}$$

Instead of considering the effect of each neighbor on a particle, we're approximating that one neighbor has nearly twice the effect on the particle.

$$Z = \left(\sum_{S_i=\pm 1} e^{\beta \mu B S_i} \right) \left(\sum_{z_i=\pm 1} e^{2\beta J z_i} \right)$$

Since there's only two possibilities, we can write the sum explicitly:

$$Z = \left(e^{\beta \mu B \cdot (+1)} + e^{\beta \mu B \cdot (-1)} \right) \left(e^{2\beta J \cdot (+1)} + e^{2\beta J \cdot (-1)} \right)$$

We recognize the form and are able to rewrite the partition function as hyperbolic cosine, again.

$$Z = 2\cosh(\beta \mu B) \cdot 2\cosh(2\beta J)$$

We also recognize the first term of this partition function to be the partition function of a paramagnetic system.

$$Z = Z_{para} 2\cosh(2\beta J)$$

For N particles, this becomes:

$$Z_N = Z_{para}^N [2\cosh(2\beta J)]^N$$

3.3 Free Energy

Similar to what we did above,

$$\begin{aligned} F &= -kT \ln[Z_{para}^N [2\cosh(2\beta J)]^N] \\ F &= -NkT \ln[Z_{para} [2\cosh(2\beta J)]] \\ F &= -NkT (\ln[Z_{para}^N] + \ln[2\cosh(2\beta J)]^N) \end{aligned}$$

This is to be expected, of course. An extra interaction is added to our model of the system.

3.4 Magnetization

For a ferromagnetic system, the magnetization can be obtained from the free energy via the thermodynamic relation:

$$M = - \left(\frac{\partial F}{\partial B} \right)_T$$

Substituting the free energy from the previous section:

$$F = -Nk_B T [\ln[2 \cosh(\beta\mu B)] + \ln[2 \cosh(2\beta J)]]$$

Only the first term depends on B , so:

$$M = Nk_B T \cdot \frac{\partial}{\partial B} \ln[2 \cosh(\beta\mu B)]$$

$$M = Nk_B T \cdot \frac{\mu\beta \sinh(\beta\mu B)}{\cosh(\beta\mu B)}$$

$$M = N\mu \tanh(\beta\mu B)$$

Thus, in the mean-field like approximation of our factorized partition function, the magnetization formula remains identical to the paramagnetic case. More sophisticated treatments (like the full 1D Ising model solution) would show spontaneous magnetization only at $T = 0$.

3.5 Magnetic Susceptibility

The magnetic susceptibility χ measures how the magnetization responds to changes in the external field:

$$\chi = \left(\frac{\partial M}{\partial B} \right)_T$$

Using $M = N\mu \tanh(\beta\mu B)$:

$$\chi = N\mu \cdot \frac{\partial}{\partial B} \tanh(\beta\mu B)$$

$$\chi = N\mu \cdot \beta\mu \operatorname{sech}^2(\beta\mu B)$$

$$\chi = N\mu^2 \beta \operatorname{sech}^2(\beta\mu B)$$

Or equivalently:

$$\chi = \frac{N\mu^2 \beta}{\cosh^2(\beta\mu B)}$$

3.6 Curie Temperature and Critical Behavior

In mean-field theory, the magnetic susceptibility near the Curie temperature T_c follows the Curie-Weiss law:

$$\frac{1}{\chi} = \frac{T - T_c}{C}$$

where C is the Curie constant. From our susceptibility expression:

$$\frac{1}{\chi} = \frac{\cosh^2(\beta\mu B)}{N\mu^2 \beta}$$

In zero field ($B = 0$), $\cosh(0) = 1$, and $\beta = 1/(k_B T)$, so:

$$\frac{1}{\chi} = \frac{k_B T}{N\mu^2}$$

Comparing with the Curie-Weiss form $\frac{1}{\chi} = \frac{T - T_c}{C}$, we would identify $C = N\mu^2/k_B$ and $T_c = 0$ for the 1D Ising model, consistent with the fact that the 1D Ising model has no finite-temperature phase transition.

Important observation: In our derived expression $1/\chi = \cosh^2(\beta\mu B)/(N\mu^2 \beta)$, the right-hand side *never diverges* for finite B and $T > 0$, because $\cosh^2(\beta\mu B)$ grows exponentially with B but remains finite, and β is finite for $T > 0$. This indicates the absence of a divergence in susceptibility at finite temperature in one dimension, except at $T = 0$ where $\beta \rightarrow \infty$.

3.7 Energy

The average energy can be obtained from the partition function via:

$$E = -\frac{\partial \ln Z_N}{\partial \beta}$$

Recall our partition function for the ferromagnetic system:

$$Z_N = [2 \cosh(\beta \mu B)]^N \cdot [2 \cosh(2\beta J)]^N$$

Taking the logarithm:

$$\ln Z_N = N \ln[2 \cosh(\beta \mu B)] + N \ln[2 \cosh(2\beta J)]$$

Now differentiating with respect to β :

$$\frac{\partial \ln Z_N}{\partial \beta} = N \cdot \frac{\mu B \sinh(\beta \mu B)}{\cosh(\beta \mu B)} + N \cdot \frac{2J \sinh(2\beta J)}{\cosh(2\beta J)}$$

$$\frac{\partial \ln Z_N}{\partial \beta} = N \mu B \tanh(\beta \mu B) + 2N J \tanh(2\beta J)$$

Therefore, the average energy is:

$$E = -\frac{\partial \ln Z_N}{\partial \beta} = -N \mu B \tanh(\beta \mu B) - 2N J \tanh(2\beta J)$$

Or explicitly in terms of temperature:

$$E = -N \mu B \tanh\left(\frac{\mu B}{k_B T}\right) - 2N J \tanh\left(\frac{2J}{k_B T}\right)$$

We can identify the first term as the energy of a purely paramagnetic system, $E_{\text{para}} = -N \mu B \tanh(\beta \mu B)$. Thus:

$$E = E_{\text{para}} - 2N J \tanh(2\beta J)$$

The negative sign in front of the second term indicates that the spin-spin interaction lowers the total energy when spins align parallel (since $J > 0$ for ferromagnetism). The $\tanh(2\beta J)$ factor represents the thermal average of the interaction energy.

3.8 Entropy

The entropy can be obtained from the thermodynamic relation:

$$S = k_B (\ln Z_N + \beta E)$$

Substituting our expressions for $\ln Z_N$ and E :

$$S = k_B [N \ln[2 \cosh(\beta \mu B)] + N \ln[2 \cosh(2\beta J)] + \beta (-N \mu B \tanh(\beta \mu B) - 2N J \tanh(2\beta J))]$$

$$S = N k_B [\ln[2 \cosh(\beta \mu B)] - \beta \mu B \tanh(\beta \mu B) + \ln[2 \cosh(2\beta J)] - 2\beta J \tanh(2\beta J)]$$

We recognize the first two terms as the entropy of a paramagnetic system:

$$S_{\text{para}} = k_B [\ln[2 \cosh(\beta \mu B)] - \beta \mu B \tanh(\beta \mu B)]$$

Thus, the total entropy can be written as:

$$S = N [S_{\text{para}} + k_B \ln[2 \cosh(2\beta J)] - 2k_B \beta J \tanh(2\beta J)]$$

Or in terms of temperature:

$$S = N k_B \left[S_{\text{para}} + \ln \left[2 \cosh \left(\frac{2J}{k_B T} \right) \right] - \frac{2J}{k_B T} \tanh \left(\frac{2J}{k_B T} \right) \right]$$

Comparison of entropy: Consider the function $f(x) = \ln(2 \cosh x) - x \tanh x$. For small x , using expansions:

$$\ln(2 \cosh x) \approx \ln 2 + \frac{x^2}{2} - \frac{x^4}{12} + \dots$$

$$x \tanh x \approx x^2 - \frac{x^4}{3} + \dots$$

Thus:

$$f(x) \approx \ln 2 - \frac{x^2}{2} + \frac{x^4}{4} + \dots$$

Since $\ln 2 > 0$, for sufficiently small x , $f(x) > 0$. This means that for small values of $2J/(k_B T)$ (i.e., high temperature or weak coupling), the additional entropy contribution from the spin-spin interactions is positive. Therefore, at high temperatures, the entropy of the ferromagnetic system is *higher* than that of a purely paramagnetic system at the same temperature and field. This makes physical sense: the spin-spin interactions introduce additional microscopic configurations and correlations that increase the disorder of the system at high temperatures.

At low temperatures ($x \rightarrow \infty$), $\ln(2 \cosh x) \approx x - \ln 2$ and $x \tanh x \approx x$, so $f(x) \approx -\ln 2 < 0$. The entropy decreases because of the ordering effect of the ferromagnetic interactions, which is consistent with the system approaching a more ordered ground state.

4 Some Mathematical Verification

While it is harder to confirm if this theory is fully mathematically correct for the 1D Ising Model's ferromagnetic applications (the Curie temperature calculated here is 0 as theoretically confirmed and that is about as much verification as I myself could find), we can better verify the paramagnetic solution.

We know $\Omega = 2^N$ for an isolated system (which is what the paramagnet is modeled as). So let us check entropy.

$$S = Nk \left[\ln \left(2 \cosh \left(\frac{\mu B}{kT} \right) \right) - \frac{\mu B}{kT} \tanh \left(\frac{\mu B}{kT} \right) \right]$$

Let $x = \mu B/kT$. Then:

$$S = Nk \left[\ln(e^x + e^{-x}) - x \cdot \frac{e^x - e^{-x}}{e^x + e^{-x}} \right]$$

For $T = 0$, we get $x \rightarrow \infty$:

$$S = Nk \left[\ln(e^\infty + e^{-\infty}) - \frac{e^\infty - e^{-\infty}}{e^\infty + e^{-\infty}} \cdot x \right]$$

$$S = Nk [\ln(e^\infty) - 1 \cdot x] = Nk [x - x] = 0$$

As $T \rightarrow \infty$, we get $x \rightarrow 0$:

$$S = Nk [\ln(e^0 + e^{-0}) - 0]$$

$$S = Nk [\ln(2)] = Nk \ln 2 = k \ln 2^N$$

The general formula is $S = k \ln \Omega$. Then $\Omega = 2^N$. We have the verification.

5 References

These notes summarize my independent re-derivation of standard Ising model results as learned in my statistical mechanics course. The Ising model was originally introduced by Ernst Ising (1925), and these thermodynamic derivations follow well-established methods in statistical physics. So any statistical mechanics textbook and even the Wikipedia pages may be referenced to confirm and reproduce what is on here. For the stuff related to magnetization and magnetic moment, defining the Hamiltonian, I referred to David Griffiths' Introduction to Electrodynamics.

I wrote these notes during my third year, struggling through the Ising model like many before me. After finally getting it, I wanted to create the guide I wish I'd had. If these notes help you, pay it forward: explain it to someone else, share your own notes, or just be patient with the next student who's struggling.